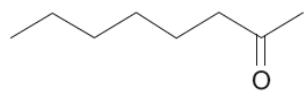
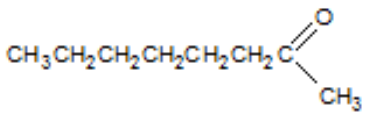
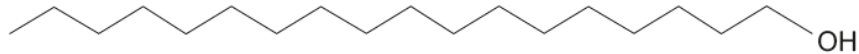


Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		$C_8H_{16}O$	1			1		
		(ii)		$NaOH(aq) / ^-OH(aq)$ (1) nucleophilic substitution (1)	2			2		1
		(iii)		accept any of following acidified dichromate(VI) acidified potassium dichromate(VI) H^+ , $Cr_2O_7^{2-}$ acidified manganate(VII) acidified potassium manganate(VII) acidified permanganate H^+ , MnO_4^-	1			1		1
	(b)	(i)		CHI_3	1			1		1
		(ii)		possible formulae  accept an unsaturated methyl secondary alcohol e.g. $CH_3CH_2CH_2CH=CHCH_2CH(OH)CH_3$ (1) methylketone / $CH_3C=O$ (1) or $CH(OH)CH_3$ (that can be oxidised to $CH_3C=O$)		1	1	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		accept any of following sodium tetrahydridoborate(III) sodium borohydride NaBH ₄ lithium tetrahydridoaluminate(III) lithium aluminium hydride LiAlH ₄		1		1		1
		(ii)		by following the disappearance of the colour using a colorimeter			1	1		1
	(d)			e.g. 99 → CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CO ⁺ / C ₅ H ₁₁ CO ⁺ / CH ₃ COCOCH ₂ CH ₂ ⁺ 71 → CH ₃ COCO ⁺ 43 → CH ₃ CO ⁺ / C ₃ H ₇ ⁺ award (2) for all three correct award (1) for any one/two correct			2	2		
	(e)	(i)		 award (1) for correct chain length award (1) for terminal primary alcohol		2		2		
		(ii)		accept any of following lithium tetrahydridoaluminate(III) lithium aluminium hydride LiAlH ₄	1			1		1
				Question 6 total	6	4	4	14	0	6